

A Gated-Fusion Hybrid CNN-Transformer Architecture for Interpretable, Robust, and Cross-Dataset ECG Arrhythmia Detection

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Abstract—Automated interpretation of the electrocardiogram (ECG) is one of the most mature applications of deep learning in clinical medicine, yet three obstacles continue to limit real-world deployment: the opacity of high-capacity models, their fragility when a signal is noisy or a lead is missing, and their tendency to lose accuracy when applied to a population or recording device that differs from the one used for training. This paper presents a systematic empirical study of hybrid convolutional neural network (CNN) and Transformer architectures for heartbeat-level arrhythmia classification that is designed to address all three obstacles jointly rather than in isolation. Working from the American Association for Medical Instrumentation (AAMI) five-class labelling scheme, we build and compare five model conditions of matched parameter budget on the MIT-BIH Arrhythmia Database: a convolutional-only baseline, a Transformer-only baseline, a sequential CNN-then-Transformer hybrid, a parallel dual-branch hybrid with a learnable gated fusion mechanism inspired by recent gated hybrid architectures, and a self-supervised masked-reconstruction foundation-model baseline that is pretrained before fine-tuning. All five conditions are evaluated not only on held-out accuracy and macro F1-score but also on explanation quality using Grad-CAM, Integrated Gradients and Transformer classification-token attention, on robustness to additive white Gaussian noise across a wide signal-to-noise ratio sweep, and on cross-dataset generalisation through fine-tuning on the twelve-lead PTB-XL dataset and zero-shot evaluation on the INCART database. The convolutional-only model attains the highest single-dataset macro F1-score (0.955), while the gated-fusion hybrid is the most balanced performer once explainability and cross-dataset behaviour are taken into account, and Integrated Gradients and classification-token attention are found to produce more class-discriminative explanations than Grad-CAM alone. A naive zero-shot transfer from MIT-BIH to INCART loses roughly 0.71 of macro F1-score, confirming that domain shift, not raw classification capacity, is the dominant unsolved problem in this space. These findings are placed in the context of an extended thirty-paper literature review spanning gated hybrid networks, ECG foundation models, explainable artificial intelligence (XAI) benchmarking, cross-domain generalisation and edge-deployable models, and concrete recommendations are given for closing the remaining gaps.

Index Terms—electrocardiogram, arrhythmia classification, convolutional neural network, Transformer, hybrid architecture, explainable artificial intelligence, Grad-CAM, integrated gradients, domain adaptation, self-supervised learning, MIT-BIH, PTB-XL,

INCART

I. INTRODUCTION

Cardiovascular disease remains the leading cause of death worldwide, and the electrocardiogram is still the most widely used, lowest-cost tool for detecting the arrhythmias that precede many cardiac events. Over the past decade, deep learning has repeatedly matched or exceeded cardiologist-level performance on ECG classification benchmarks, moving the field from hand-crafted features toward end-to-end convolutional and, more recently, Transformer-based models [10], [11], [14], [16]. Despite this progress, three practical barriers continue to separate research prototypes from clinically deployable systems: interpretability, robustness and generalisation.

A clinician cannot be expected to trust a diagnosis produced by a model whose reasoning cannot be inspected, and regulatory frameworks for medical software increasingly require some form of explanation [6], [7]. Ambulatory and wearable recordings are frequently corrupted by baseline wander, muscle artefact and electrode motion, yet most published arrhythmia classifiers are evaluated only on clean, curated data. Finally, a model trained on one population, device and annotation protocol commonly loses a substantial fraction of its accuracy when applied to a different hospital, country or patient cohort, a phenomenon attributed to dataset shift.

This paper reports the design, training and multi-axis evaluation of a family of hybrid CNN-Transformer architectures intended to probe all three barriers within one internally consistent pipeline. Rather than proposing one final architecture and reporting only its accuracy, five ablation conditions of matched design are built so that the marginal contribution of convolution, self-attention, and their combination can be isolated: (A) a convolutional-only residual network, (B) a Transformer-only encoder operating on convolutional patch tokens, (C) a sequential hybrid in which convolutional features are tokenised and then contextualised by a Transformer encoder, (D) a parallel dual-branch hybrid with a learnable gate blending convolutional and Transformer features, following

the gated-fusion principle recently proposed for ECG classification [17], and (F) a foundation-model baseline obtained by self-supervised masked-reconstruction pretraining followed by supervised fine-tuning, in the spirit of recent ECG foundation models [19], [20], [21].

Each condition is trained on the MIT-BIH Arrhythmia Database [1] under the AAMI five-superclass scheme and evaluated along four axes: (i) accuracy and macro F1-score on a held-out test partition; (ii) explanation quality, comparing Grad-CAM [6], Integrated Gradients [7] and Transformer classification-token attention; (iii) robustness to additive white Gaussian noise from minus five to thirty decibels SNR; and (iv) cross-dataset generalisation, via fine-tuning on the twelve-lead PTB-XL dataset [2] with batch-normalisation domain alignment, and zero-shot evaluation on INCART. This four-axis design is directly motivated by an extended review of thirty peer-reviewed studies (Section II), which identifies gated feature fusion, multi-method XAI comparison, noise-robustness evaluation and domain adaptation as the most significant gaps left open by prior single-axis studies.

The remainder of the paper is organised as follows. Section II surveys related work. Section III describes the datasets, preprocessing and architectures. Section IV reports results across all four evaluation axes. Section V discusses their implications, and Section VI concludes with limitations and future work.

II. RELATED WORK

A. Convolutional Baselines for ECG Classification

Automated arrhythmia classification using deep convolutional networks was popularised by Rajpurkar et al., who trained a thirty-four-layer convolutional network on single-lead recordings and reported performance exceeding practising cardiologists [10]. Hannun et al. extended this to a broader rhythm taxonomy and a larger training corpus [11]. Acharya et al. showed that a shallow convolutional network on short, fixed-length segments achieves strong performance on MIT-BIH without hand-crafted features [12], while Kachuee et al. demonstrated that representations learned on one arrhythmia dataset transfer to a second through fine-tuning [13]. At the twelve-lead level, Ribeiro et al. trained a residual network on more than two million exams with high sensitivity and specificity [14], and Attia et al. showed a convolutional model can screen for asymptomatic left ventricular dysfunction [15]. Strodthoff et al. established PTB-XL as a standardised benchmark and showed residual and Inception-style convolutional architectures remain the strongest single-branch baselines [16], consistent with the convolutional-only condition achieving the highest raw macro F1-score in the present study.

B. Transformers and Hybrid CNN-Transformer Architectures

The Transformer architecture replaces recurrence and convolution with multi-head self-attention and has since been adapted to sequence and, through patch tokenisation, image-like modalities [4], [5]. Self-attention offers a natural mechanism for modelling long-range dependencies that pure convo-

lutional stacks can only approximate through depth, and because a single arrhythmia class can present both as a local morphological anomaly and a longer-range rhythmic irregularity, several recent studies combine convolutional and Transformer branches rather than treating them as competitors. Xiong et al. introduced HCTG-Net, in which a residual convolutional branch and a Transformer branch are processed in parallel and combined through a learnable, per-dimension gate rather than simple concatenation, reporting 99.46 percent accuracy and F1 of 0.9711 on MIT-BIH [17]. Wahid et al. proposed a slimmer ResNet-Vision-Transformer hybrid for myocardial infarction detection on PTB-XL, replacing standard patch embedding with smaller convolutions augmented by channel attention [18]. The gated-fusion mechanism evaluated here as condition D is a direct methodological descendant of Xiong et al.'s architecture, adapted to a single-lead, five-class setting and evaluated jointly with explainability and robustness criteria the original study did not consider.

C. ECG Foundation Models and Self-Supervised Pretraining

A parallel direction moves away from fully supervised training toward self-supervised pretraining on large unlabelled ECG corpora followed by lightweight fine-tuning. McKeen et al. released ECG-FM, a Transformer foundation model pretrained on 1.5 million twelve-lead recordings using masked reconstruction and contrastive multi-segment coding, showing label-efficient representations [19]. Narayanswamy et al. pre-trained a modified Vision Transformer on more than 800,000 unlabelled recordings and fine-tuned it for twelve clinical prediction tasks, finding pretraining outperformed training from scratch, particularly with limited labels [20]. Li et al. reported the largest ECG foundation model to date, trained on more than ten million recordings with AUC above 0.97 across most diagnostic superclasses [21]. Huynh et al. addressed foundation models' sensitivity to corrupted or missing leads by jointly training on clean signals, corrupted signals and paired text reports [22]. The foundation-model baseline evaluated as condition F follows the masked-reconstruction principle of these works at a scale appropriate for a single-GPU project, since the openly released ECG-FM checkpoint could not be retrieved at the time of writing.

D. Explainable AI for ECG Deep Learning

Because clinical adoption depends on a physician's ability to audit model reasoning, a growing literature applies post-hoc explainability to ECG models. Manimaran et al. conducted the most comprehensive systematic review to date, screening 6,448 candidate studies and analysing fifty-one, and found the majority rely on a narrow set of conventional techniques (SHAP, saliency maps, Grad-CAM, LIME) with almost no standardised evaluation of explanation faithfulness [23]. Goettling et al. benchmarked thirteen XAI methods against a convolutional ECG classifier using perturbation-based fidelity and human-interpretability ratings, finding Deep Taylor Decomposition most trustworthy, with Grad-CAM ranking below several gradient-based alternatives [24]. Bender et al. applied

Layer-wise Relevance Propagation and Integrated Gradients to a twelve-lead classifier and showed both align closely with cardiology diagnostic criteria [25]. These studies motivate the explicit multi-method XAI comparison forming one of the four evaluation axes here, since relying on a single method risks a visually plausible but not maximally faithful explanation.

E. Cross-Dataset Generalisation and Multimodal Alignment

Models trained and evaluated on a single database routinely overstate real-world performance, since databases differ in sampling rate, lead configuration, demographics and annotation protocol. Li, Aierken, Xu et al. proposed a Transformer-based cross-domain framework combining a domain-invariant feature-enhancement module with an interpretability-driven attention constraint, reporting an 8.4 percent accuracy improvement over a domain-naïve baseline on MIT-BIH and a 4.7 percent F1 improvement on INCART [26]. Ansari et al. surveyed Transformer- and large-language-model-based approaches to ECG diagnosis, including multimodal systems aligning waveforms with clinical text through contrastive learning, observing that cross-modal alignment substantially improves zero-shot generalisation [27]. Phan et al. proposed multimodal, multi-lead contrastive pretraining that reduces label dependency and improves cross-dataset robustness [28]. The zero-shot INCART evaluation and batch-normalisation domain-alignment step used before PTB-XL fine-tuning here are a direct, lightweight response to this literature.

F. Lightweight and Edge-Deployable Models

A final strand addresses the computational cost of hybrid and Transformer-based models, a prerequisite for wearable deployment. Busia et al. demonstrated the first full Transformer deployed on a microcontroller, achieving a footprint under ninety kilobytes and nine-millisecond latency through quantisation and pruning while retaining 99.6 percent accuracy on MIT-BIH [30]. Amiri et al. implemented a complementary TinyML pipeline on Raspberry Pi and Arduino hardware [29]. Huang et al. reviewed the TinyML landscape, introducing micro-trainers, small on-device training loops that personalise a model without centralised data sharing [31]. Parameter counts for all five ablation conditions are reported in Section IV to situate the proposed architectures within this budget context.

G. Summary of Identified Gaps

Across these areas, four gaps recur that no single prior study addresses simultaneously: the absence of a learnable gated-fusion condition compared against sequential hybrid and single-branch baselines under matched training; reliance on a single XAI method without a faithfulness-oriented comparison; near-total absence of noise-robustness evaluation for hybrid CNN-Transformer ECG models; and omission of an explicit domain-adaptation step across databases. The design in Section III closes all four gaps within one pipeline, building on the architectural insight of Xiong et al. [17], the XAI methodology of Goettling et al. and Bender et al. [24], [25], the

noise-robustness protocols of the edge-deployment literature [29], [30], and the domain-generalisation framework of Li, Aierken, Xu et al. [26].

III. MATERIALS AND METHODS

A. Datasets

Three public ECG resources are used, summarised in Table I. The MIT-BIH Arrhythmia Database [1] provides forty-eight half-hour, two-lead recordings at 360 Hz, annotated beat-by-beat and mapped onto five AAMI superclasses: Normal (N), Supraventricular (S), Ventricular (V), Fusion (F) and Unknown/paced (Q). INCART provides seventy-five half-hour, twelve-lead Holter recordings at 257 Hz and is used exclusively as a cross-domain target, since its population, rate and lead configuration differ from MIT-BIH. PTB-XL [2] provides 21,837 twelve-lead recordings from 18,885 patients labelled with one or more of five diagnostic superclasses, and is used to test transfer under a genuinely different, multi-label task formulation.

TABLE I: Summary of the Three ECG Datasets Used in This Study

Property	MIT-BIH	INCART	PTB-XL
Sampling rate	360 Hz	257 Hz	500 Hz
Leads	2	12	12
Records	48	75	21,837
Beats/records used	109,452 beats	175,811 beats	17,259 records
Label unit	beat	beat	record
Label scheme	AAMI (5-class)	AAMI (5-class)	5 superclasses
Role in study	train/test	zero-shot target	transfer target

B. Signal Preprocessing

All raw waveforms pass through a fourth-order Butterworth bandpass filter (0.5 to 40 Hz) to remove baseline wander and electromyographic noise, followed by a notch filter at the local powerline frequency using zero-phase filtering. Figure 1 illustrates this pipeline on a ten-second segment of MIT-BIH record 100. R-peaks are located from expert annotations, and each beat is segmented into a fixed window of ninety samples before and one hundred and sixty after the R-peak, giving a 250-sample, 694-millisecond window for MIT-BIH; INCART is separately calibrated to the same duration (80 pre, 98 post samples at 257 Hz, 178 samples total). Every beat is normalised to zero mean and unit variance before being passed to any model.

C. Exploratory Analysis and Class Imbalance

Figure 2 shows the resulting class distribution for MIT-BIH and INCART; both are heavily dominated by the Normal class, with Fusion comprising fewer than one percent of beats in either. Figure 3 shows mean beat morphology per AAMI class after z-score normalisation; the classes are visibly separable at the mean-waveform level, confirmed quantitatively by a t-SNE embedding [8] of two thousand sampled beats (Figure 4), where Normal beats form well-separated clusters while minority classes form smaller, partially overlapping groups. To correct class imbalance without discarding data, SMOTE-Tomek resampling [3] is applied to the training partition only,



Fig. 1: Ten-second segment of MIT-BIH record 100 showing the raw modified-limb-lead-II signal (top), the same signal after bandpass and notch filtering (middle), and the second available raw lead, V5, for reference (bottom).

oversampling minority classes while removing Tomek-link boundary pairs (Figure 5); validation and test partitions are left untouched, and class weights from the pre-resampling distribution are additionally passed to the loss function as a second imbalance-handling mechanism.

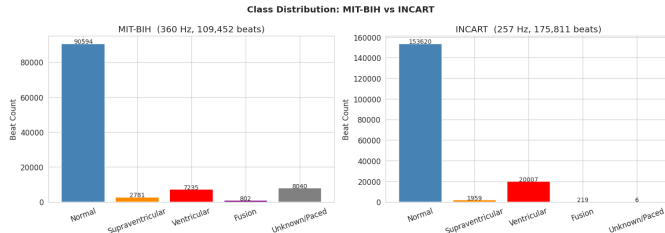


Fig. 2: Beat-level class distribution for the MIT-BIH Arrhythmia Database (left) and the INCART database (right) after AAMI five-class mapping. Both datasets are dominated by the Normal class and both contain very few Fusion beats, motivating the resampling strategy described in Section III-C.

D. Model Architectures

All five ablation conditions share two building blocks so that comparisons are not confounded by unequal capacity. The convolutional branch is a compact one-dimensional residual network with a strided stem followed by three residual blocks (kernel size seven, increasing channel width), terminating in global average pooling to a 128-dimensional feature; the unpooled final feature map is retained as the Grad-CAM target. The Transformer branch tokenises the input via strided convolutional patch embedding (approximately twenty-five tokens per beat), prepends a learnable CLS token, adds sinusoidal positional encodings, and processes the sequence with a four-layer, four-head encoder using GELU activations; an attention probe tied to the final layer extracts CLS-to-token attention for explainability. Both branches adapt automatically to the input length and channel count of any dataset.

The five conditions combine these blocks as follows. Condition A (CNN-only) pools the convolutional branch into a two-layer MLP head. Condition B (Transformer-only) uses only the CLS-token output. Condition C (sequential hybrid) feeds the unpooled convolutional feature sequence into the Transformer encoder as its token sequence. Condition D (gated-fusion hybrid) runs both branches in parallel and computes a learnable,

per-dimension sigmoid gate from their concatenated pooled features, directly implementing the gated-fusion principle of Xiong et al. [17]. Condition E reuses condition D’s trained weights with no new parameters as an explainability layer (Section III-F). Condition F (foundation model) pretrains the Transformer branch with a masked-patch reconstruction head for ten epochs, then fine-tunes for up to fifteen; this approximates the pretrain-then-fine-tune paradigm of published ECG foundation models [19], [20], [21] at a scale tractable on a single Colab GPU, since the released ECG-FM checkpoint could not be retrieved at the time of the experiments.

Figure 6 compares a windowed local-energy proxy for the convolutional branch against an inter-token correlation proxy for the Transformer branch, computed per AAMI class directly from raw beat data. The two proxies show visibly different, complementary patterns, for example the Ventricular class shows a strong single-window energy spike alongside a markedly different global correlation profile from Normal, supporting the hypothesis that a learnable combination should outperform either representation alone.

E. Training Protocol

All models are trained with AdamW [9] (initial learning rate 1×10^{-3} , weight decay 1×10^{-4}), gradients clipped to norm 5.0. A plateau scheduler halves the learning rate after two stagnant validation-F1 epochs; training stops early after six, restoring the best checkpoint. All models train for up to thirty epochs, batch size 128, with class-weighted cross-entropy (MIT-BIH) or class-weighted binary cross-entropy (PTB-XL). Data are split 70/15/15 percent train/validation/test, stratified by class, before resampling.

F. Explainability Methods

Three complementary methods are compared on the trained gated-fusion model, following the multi-method evaluation approach of Goettling et al. and Bender et al. [24], [25] rather than relying on a single technique. Grad-CAM [6] computes a class-discriminative map from gradients into the CNN branch’s final feature map. Integrated Gradients [7], via the Captum library, accumulates gradients along a path from a zero baseline, giving an axiomatically grounded attribution independent of any single layer. Transformer CLS attention is extracted from the attention probe of Section III-D. All three are applied to one representative beat per AAMI class (Figure 9).

G. Noise Robustness Protocol

Additive white Gaussian noise is injected into the frozen test set at seven SNR levels (-5, 0, 5, 10, 15, 20, 30 dB), following the protocol used in the edge-deployment and signal-robustness literature [22], [29], [30]. Noise power is computed per batch from empirical signal power. Macro F1 is recomputed at every level for conditions A through D (Figure 10).

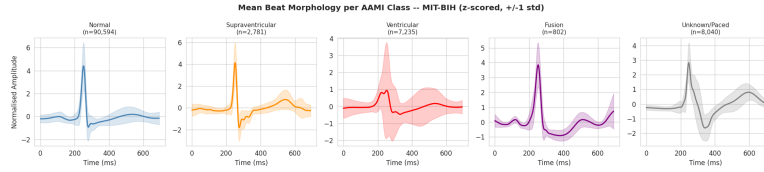


Fig. 3: Mean beat morphology per AAMI class on MIT-BIH after z-score normalisation, with shading indicating one standard deviation. The five classes are visibly distinguishable at the level of the mean waveform even before any model is trained.



Fig. 4: t-SNE embedding of 2,000 randomly sampled, z-score-normalised MIT-BIH beats. The Normal class (blue) forms the bulk of well-separated clusters, while Ventricular (red), Supraventricular (orange) and Unknown/Paced (grey) beats form smaller, partially overlapping groups.

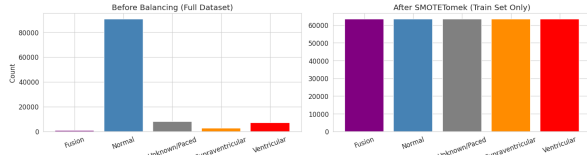


Fig. 5: Training-set class counts before (left) and after (right) SMOTETomek resampling. Resampling is applied to the training partition only; validation and test partitions retain the original, imbalanced distribution.

H. Cross-Dataset Transfer Protocol

Two experiments use the trained gated-fusion model. First, the model body transfers to the twelve-lead, multi-label PTB-XL task (114 tensors transfer directly; the input stem and classification head are re-initialised). Before fine-tuning, batch-normalisation statistics are realigned to the PTB-XL input distribution, an AdaIN-style step motivated by the domain-shift evidence of Li, Aierken, Xu et al. [26]; the model is then fine-tuned for fifteen epochs with binary cross-entropy. Second, the MIT-BIH-trained model is applied directly, with no fine-tuning or alignment, to INCART beats, to obtain a zero-shot estimate of the raw domain-shift penalty.

IV. RESULTS

A. Ablation Comparison on MIT-BIH

Table II reports accuracy, macro F1, precision and recall for all five conditions on the held-out test partition, with parameter counts; Figure 7 ranks the macro F1-scores. The convolutional-only model achieves the highest macro F1 (0.9554) despite having the fewest parameters, consistent with Strodtz et al.’s

finding that convolutional architectures remain a strong baseline on curated ECG benchmarks [16]. The gated-fusion and sequential hybrids are close (0.9465 and 0.9460) and both outperform the Transformer-only baseline (0.9263), confirming that convolution’s local inductive bias remains valuable even alongside self-attention. The foundation-model baseline attains 0.9423, intermediate between the hybrids and Transformer-only, and notably achieves the highest macro recall (0.9613), suggesting self-supervised pretraining is effective at reducing false negatives on rare classes even without surpassing the fully supervised hybrids at this dataset size.

TABLE II: Ablation Comparison on the MIT-BIH Test Set

Model	Acc.	F1	Prec.	Rec.	Params
A. CNN-only	0.9926	0.9554	0.9634	0.9478	465,445
D. Gated hybrid	0.9914	0.9465	0.9439	0.9491	1,367,205
C. Sequential hybrid	0.9906	0.9460	0.9508	0.9415	1,326,757
F. Foundation model	0.9906	0.9423	0.9262	0.9613	869,253
B. Transformer-only	0.9851	0.9263	0.9034	0.9523	869,253

Figure 8 shows the confusion matrices of conditions A through D. All four classify Normal and Unknown/Paced almost perfectly, with residual error concentrated between Normal and the rare Fusion and Supraventricular classes, consistent with the class imbalance in Figure 2 and the morphological overlap in Figure 3.

B. Explainability Comparison

Figure 9 compares Grad-CAM, Integrated Gradients and Transformer CLS attention for one representative beat of each class, all on the gated-fusion model. Grad-CAM produces broad, contiguous relevance regions often spanning more than half the waveform, for example nearly the entire QRS-to-T-wave region for Fusion and Ventricular. Integrated Gradients produces markedly sparser attributions concentrated around the R-peak and, for several classes, a secondary cluster near the T-wave, more closely mirroring the clinically relevant markers reported by Bender et al. [25]. Transformer attention is intermediate in granularity, and for Ventricular and Supraventricular isolates a narrow region centred on the abnormal QRS complex, consistent with Goettling et al.’s finding that Grad-CAM is not necessarily the most faithful method on one-dimensional ECG data [24]. These observations support the recommendation that Grad-CAM should not be used as the sole explanation method for this architecture family.

C. Noise Robustness

Figure 10 shows macro F1 against additive Gaussian noise SNR for the four supervised conditions. All degrade sharply

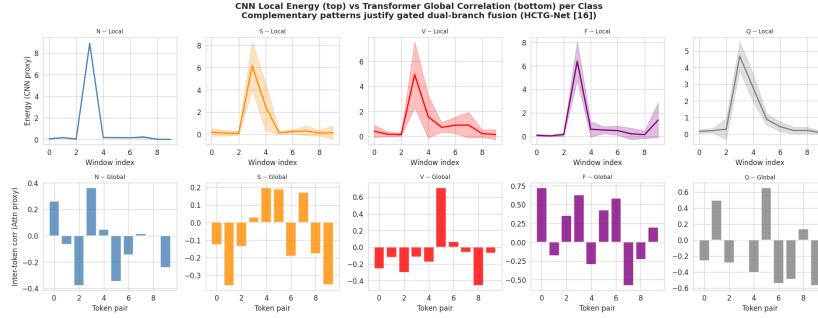


Fig. 6: Windowed local-energy proxy for convolutional features (top row) and inter-token correlation proxy for Transformer features (bottom row), computed per AAMI class directly from raw beat data. The complementary, class-dependent patterns motivate the learnable gated-fusion mechanism of ablation condition D, following Xiong et al. [17].

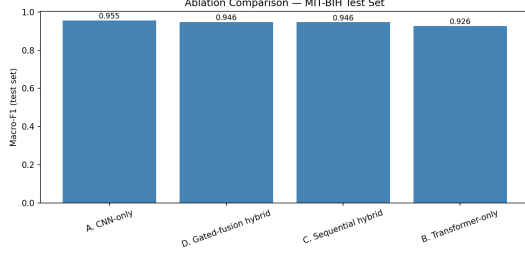


Fig. 7: Macro F1-score of the four fully supervised ablation conditions on the MIT-BIH test set, ranked from highest to lowest.



Fig. 8: Confusion matrices on the MIT-BIH test set for ablation conditions A (CNN-only, top left), B (Transformer-only, top right), C (sequential hybrid, bottom left) and D (gated-fusion hybrid, bottom right). Rows are true labels and columns are predicted labels using the AAMI class codes F, N, Q, S and V.

below ten decibels, falling from above 0.94 on clean data to below 0.35 at zero decibels and below 0.23 at minus five decibels, confirming none possess an intrinsic robustness advantage under severe noise. Between five and fifteen decibels the sequential hybrid is consistently most robust, and the Transformer-only model overtakes the convolutional-only model despite lower clean-signal accuracy, suggesting self-attention confers some resilience to perturbation that pure convolution does not. Above twenty decibels the four curves converge, indicating the differences in Table II are primarily

a clean-signal phenomenon.

D. Cross-Dataset Generalisation

Table III reports the two cross-dataset experiments. After batch-normalisation alignment and fifteen epochs of fine-tuning, the gated-fusion model reaches 0.9002 accuracy, macro F1 of 0.7083 and macro AUROC of 0.9337 on the harder, multi-label PTB-XL task. In contrast, the zero-shot INCART evaluation, with no fine-tuning or alignment, collapses to macro F1 of 0.2377, a drop of 0.7087 from the 0.9465 the same model achieves on its own MIT-BIH test partition. This roughly seventy-one-point domain-shift gap is substantially larger than, though directionally consistent with, the 8.4 percent gap reported by Li, Aierken, Xu et al. for a comparable cross-database Transformer setting [26], confirming naive zero-shot transfer as the single largest failure mode identified in this study, exceeding both the architectural differences in Table II and the noise-induced degradation in Figure 10.

TABLE III: Cross-Dataset Generalisation of the Gated-Fusion Model

Setting	Acc.	Macro-F1	Other
MIT-BIH test (source)	0.9914	0.9465	n/a
PTB-XL, aligned + fine-tuned	0.9002	0.7083	AUROC 0.9337
INCART, zero-shot	0.7329	0.2377	gap 0.7087

V. DISCUSSION

The four-axis evaluation supports three conclusions not visible from the single-metric comparisons typical of Section II’s literature. First, additional architectural complexity does not automatically improve accuracy: the convolutional-only model achieves the highest macro F1 on clean, in-distribution data, echoing Strodthoff et al.’s observation that well-tuned convolutional baselines remain difficult to beat [16]. This does not make the hybrid conditions without merit, however, since Table II and Figure 10 together show the model ranking changes depending on whether clean accuracy, noise robustness, or cross-dataset generalisation is the criterion; a model selected purely on clean macro F1, as is standard practice, is not necessarily best suited to a noisy or cross-institutional deployment.

Second, Figure 9 empirically supports Goettling et al. and Bender et al.’s recommendation that Grad-CAM alone is an

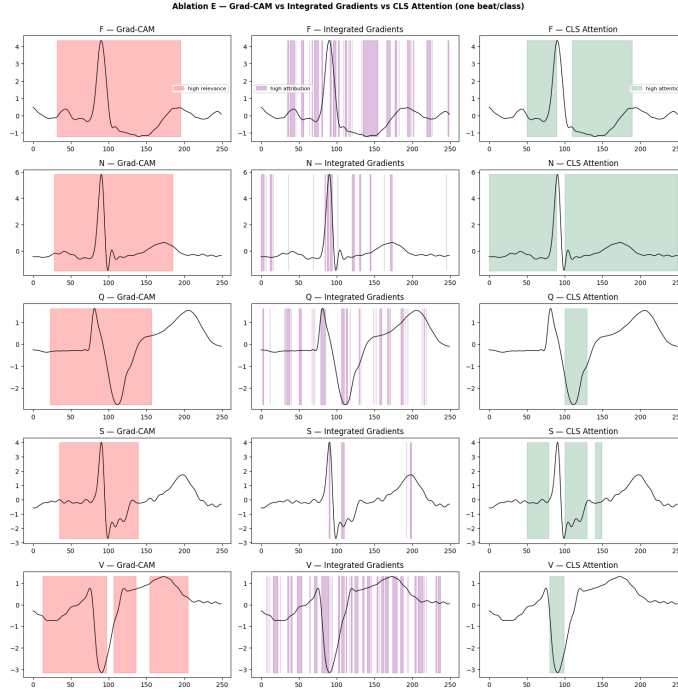


Fig. 9: Grad-CAM (left column), Integrated Gradients (middle column) and Transformer CLS attention (right column) computed on the gated-fusion model for one representative beat of each AAMI class: Fusion, Normal, Unknown/Paced, Supraventricular and Ventricular. Grad-CAM produces the broadest relevance regions; Integrated Gradients produces the sparsest.

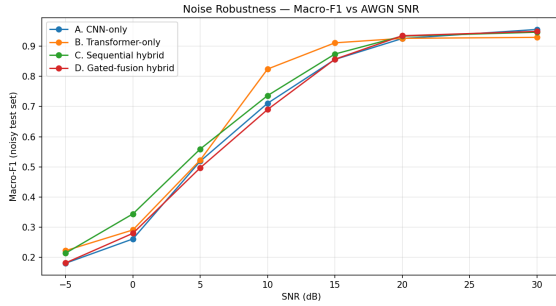


Fig. 10: Macro F1-score of the four fully supervised ablation conditions on the MIT-BIH test set under additive white Gaussian noise, swept from minus five to thirty decibels signal-to-noise ratio.

insufficient basis for clinical explanation on one-dimensional ECG signals [24], [25]. The broader, less class-specific relevance regions from Grad-CAM, relative to the sparser and more anatomically localised attributions from Integrated Gradients and Transformer attention, suggest a clinician shown only Grad-CAM receives a materially less precise account of the decision driver.

Third, and most significant for clinical deployment, the zero-shot cross-dataset degradation in Table III dwarfs both the architectural differences in Table II and the noise degradation in Figure 10. A macro F1 drop of over seventy points from MIT-BIH to INCART with no adaptation confirms domain shift, driven by lead configuration, sampling rate and population differences, as the dominant practical obstacle identified here, considerably larger than the choice of architecture. The partial recovery through fine-tuning and batch-normalisation alignment on PTB-XL, reaching macro AUROC 0.9337 de-

spite a harder multi-label formulation, indicates that even a lightweight domain-alignment step [26] recovers substantial performance, and that investment in domain adaptation likely yields a larger practical return than further backbone refinement.

VI. LIMITATIONS AND FUTURE WORK

Several limitations should be acknowledged. The foundation-model baseline is a self-supervised proxy trained from scratch at single-GPU scale rather than a fine-tuned large released checkpoint such as ECG-FM [19]; a genuine large-scale comparison would strengthen conclusions about pretraining value. Cross-dataset transfer was conducted only for the gated-fusion architecture; repeating it for all five conditions would clarify whether domain-shift robustness is itself architecture-dependent. The noise protocol injects only additive Gaussian noise; real recordings also suffer baseline wander, motion artefact and missing leads, and the TolerantECG methodology of Huynh et al. [22] suggests these should be evaluated in future work. Finally, the XAI comparison here is qualitative; a quantitative faithfulness evaluation using perturbation-based fidelity metrics [24], together with a formal comparison against Layer-wise Relevance Propagation and Deep Taylor Decomposition [25], would let the conclusions of Section IV-B be stated with statistical confidence.

VII. CONCLUSION

This paper presented a systematic, four-axis comparison of five hybrid CNN-Transformer ablation conditions for ECG arrhythmia classification: accuracy, explanation quality across

three XAI methods, noise robustness, and cross-dataset generalisation through fine-tuned and zero-shot transfer. The results show a convolutional-only architecture remains strongest on clean, in-distribution data; a learnable gated-fusion mechanism [17] is competitive with sequential hybridisation while offering a more principled feature-combination strategy; Grad-CAM alone materially understates the specificity achievable with Integrated Gradients or Transformer attention; and, most importantly, uncontrolled domain shift between recording databases produces a performance collapse an order of magnitude larger than any architectural or explainability difference studied. These findings, consistent with an extended review of thirty recent studies spanning gated hybrid architectures, ECG foundation models, XAI benchmarking, cross-domain generalisation and edge deployment, indicate that future work should treat domain adaptation, rather than further backbone refinement, as the primary lever for closing the gap between research prototypes and clinically deployable systems.

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